

SERVANDO AVALOS

Cambridge, MA | Houston, TX | Portfolio: [MechE-Servando.com](https://www.mechE-servando.com)

EDUCATION

Massachusetts Institute of Technology

Bachelor of Science in Mechanical Engineering (Product Design Concentration)

Cambridge, MA

May 2026 | GPA: 4.6/5.0

RELEVANT EXPERIENCE

MIT LMP

Fall 2025

2.008 Lab Assistant

- Guided 2 student teams in iterating and manufacturing 50 Yo-Yos each in a fast-paced shop space.
- Assisted in SLA printing thermoforming molds for 12 student teams across the entire 2.008 course.
- Identified and troubleshooted issues in manufacturing processes for teams on injection molders, CNC mills, and thermoformers.

Axon Enterprise

Summer 2025

Hardware Engineering Intern

- Designed and iterated an acrylic test fixture for an early-stage body-worn camera prototype, enabling efficient hardware debugging and testing across 50 global electrical and software teams.
- Traveled to Taiwan to support on-site prototype assembly of an unreleased body-worn camera, directly assisting in physical builds, documenting process feedback, and functional testing during the ME Proto phase.
- Conducted mechanical tolerance analysis and top-down CAD modeling using Creo, optimizing component fit and foam compression within multi-material enclosures in dense consumer electronic assemblies.

KULR Technology

Summer 2024

Prototyping and Testing Engineering Intern

- Led the research and design of a BB-2590 lithium-ion battery pack on SolidWorks based on military standards using injection molding design techniques.
- Assisted with manufacturing and assembly processes for space and defense battery packs in a fast-paced startup environment.

MIT Arcturus Autonomous Robotics

July 2023 – February 2025

MechE Subteam / Media Lead

- Planning the strategic design of a watertight command module using OnShape to meet custom electrical requirements.
- Implemented strategic posts resulting in a 34% follower increase (200+ followers) on social media platforms.

MIT Laboratory for Aviation and the Environment

September 2023 – February 2024

Undergraduate Researcher

- Manufactured a single-stage thruster integral in advancing fixed-wing ionic aircraft propulsion research.
- Designed and prototyped enhanced components using SolidWorks, strategically addressing critical failure points in the thruster to improve overall structural reliability.

RELEVANT COURSEWORK

Product Engineering Process (2.009)

Fall 2025

- Worked in a team of 15 to design and construct a high-quality product prototype while integrating user feedback.

Design of Compliant Mechanisms, Machines, & Systems (2.145)

Fall 2025

- Designed and fabricated 5 compliant mechanism projects in a team of 6 after setting deliberate functional requirements.

Medical Device Design (2.750)

Spring 2025

- Developed a low-cost RFID system for ambulation tracking that collects walking data for post-operative patients in medical-surgical wards with high nurse-to-patient ratios.

SKILLS

Software: Creo | SolidWorks | Fusion 360 | Python | Arduino IDE | MATLAB | CAD | CAM

Hardware: 3D Printing – FDM, SLS, SLA | Mill | Lathe | Waterjet | CNC | Laser Cutter

Other: Fluent in Spanish